

Guide to this Project

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This guide accompanies the article. It maps its numbered statements to proof locations, compares the cited and implemented proof methods, and indexes reusable classical results. The article’s Section 2 fixes the analytic framework; Sections 3 and 4 treat the torus and whole space. Source locations refer to the accompanying repository snapshot.

1 Build requirements

Use Lean 4.34.0-rc2 and the checked-in manifests, which pin Mathlib to 85e3a25e006c35636f0e53b0e9296caca2685bc0. Build and test in Linux:

```
lake -d verification exe cache get
make check
make test
make test-mutations
```

The suite checks 30 current publication interfaces. The separate declaration audit checks the transitive axioms of the results cited in this guide. For macOS 27, use a Linux VM because of the pinned runtime’s allocator issue. The repository README gives the build details.

2 Finding the theorem corresponding to the article

Each entry gives the proof file, declaration line and theorem name. The prefixes F/ and B/ mean `formalization/NSFormalization/` and `verification/Bindings/`, respectively. Names are local to the indicated file’s namespace; line numbers refer to the source tree identified above. A `def` entry is explicitly a constructed witness, not a theorem declaration. Some final proofs assemble several results in `Bindings`; these are distinct from the conformance wrappers in `verification/Tests/`.

Closed means that the complete article statement has a Lean kernel-checked proof, including every auxiliary result and its instantiation. No unproved theorem input, `sorry/admit`, or extra axiom is allowed. Only the article’s stated hypotheses and the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound` remain. **Partial** means that only part of the article statement is formally proved. A downstream result can be Closed when it uses only fully formalized cases of a Partial result.

All 26 numbered statements and the numbered remark appear below. The table follows the article’s numbering; only its two main theorems retain parenthetical titles. Qualifications record differences in scope. Historical identifiers retain *insertion* and *packet* where the article uses *gluing* and *building block*.

Article statement	Proof location and scope
Theorem 1.1	Closed. F/Source/PacketBreakdown.lean:76, theorem source_breakdown. Construction and same-force global nonexistence for every positive viscosity, with blowup time fixed at one.
Proposition 2.1	Partial. B/LocalTheoryV2.lean:93, def localTheoryV2; F/Section4/A02/Uniqueness.lean:120, theorem velocity_unique; F/Section4/A02/Maximal.lean:157, theorem exists_maximal_of_localSolution; F/Section4/A04/ShiftedExtension.lean:247, theorem extendsBeyond_of_memForceR'; B/TorusLocalTheory.lean:483, def torusLocalTheory. Proved for the density force classes, with H^7/H^3 restart. The general H^1 -uniform restart statements are not proved (Section 3).
Lemma 2.2	Closed. F/Section3/T14/PacketEnergy.lean:160, theorem energy_le_work_of_packet; F/Section3/T14/PacketEnergy.lean:106, theorem work_eq_square_of_packet; F/Section4/I01/Quiet.lean:53, theorem exists_packet_quiet.
Theorem 3.1	Closed. F/Section3/T21/MainAssembly.lean:84, theorem mainStatement_holds.
Lemma 3.2	Closed. F/Section3/T13/Assembly.lean:330, theorem localizationAPI. Includes the interior-order estimate and the two endpoint identities.
Proposition 3.3	Closed. F/Section3/T15/Assembly.lean:130, theorem scalingStatement_holds.
Lemma 3.4	Closed. F/Section3/T16/Assembly.lean:483, theorem localPotential.
Lemma 3.5	Partial. F/Section3/T17/Assembly.lean:107, theorem correctionStatementAmended_holds. The formal statement includes explicit smoothness, chart and positive-viscosity hypotheses. These are supplied in the downstream gluing theorem.
Theorem 3.6	Partial. F/Section3/T18/Assembly.lean:238, theorem periodicInsertionStatement_holds; F/Section3/T19/Threading.lean:126, def insertion. The generic assembly takes scaling and correction records. The construction from raw data uses a fixed ball; the prescribed-ball version is not yet closed.
Proposition 3.7	Closed. F/Section3/T19/Assembly.lean:43, theorem periodicDensityStatement_holds.
Proposition 3.8	Closed. F/Section3/T20/Assembly.lean:53, theorem criticalRegularityStatement_holds.
Corollary 3.9	Closed. F/Section3/T21/MainAssembly.lean:80, theorem nonDensityStatement_unconditional.
Corollary 3.10	Closed. F/Section3/T19/Assembly.lean:47, theorem mixedRegionStatement_holds.
Corollary 3.11	Closed. F/Section3/T19/Assembly.lean:51, theorem strongClosureStatement_holds.
Proposition 3.12	Closed. F/Section3/T19/Assembly.lean:55, theorem projectionStatement_holds.
Remark 3.13	Closed. F/Section3/T18/ForceAmplitude.lean:24, theorem packetForce_ne_zero; F/Section3/T18/ForceAmplitude.lean:240, theorem forceAmplitude_lower; F/Section3/T18/ForceAmplitude.lean:288, theorem forceAmplitude_diverges; F/Section3/T18/ForceAmplitude.lean:332, theorem forceAmplitude_real_diverges; B/ForceAmplitude.lean:40, theorem forceAmplitude_from_data. The positive finite packet amplitude gives the displayed lower bound and divergence for every inserted family.
Corollary 3.14	Closed. B/BoundaryInsertionV2.lean:27, theorem boundaryInsertion_from_data; F/Section3/T22/Assembly.lean:25, theorem boundedDomainNorm.

Article statement	Proof location and scope (continued)
Proposition 3.15	Closed. B/AffineVariation.lean:109, theorem affineVariationPacket. Instantiated at the selected upstream building block.
Proposition 3.16	Partial. F/Section3/T24/MultipleAssembly.lean:51, theorem multipleRegionsStatement_holds. Periodic case; the bounded-domain variant is not included.
Proposition 3.17	Partial. F/Section3/T24/ConservativeAssembly.lean:45, theorem conservativeForcing. Periodic case; the bounded-domain variant is not included.
Theorem 4.1	Closed. B/MainThresholds.lean:85, theorem mainThresholds. Density: B/DensityFromInsertion.lean:30, theorem breakdownDenseR_of_subcritical; converse: B/MainThresholds.lean:34, theorem mainThresholds_nonDensity.
Theorem 4.2	Closed. B/InsertionFromData.lean:150, theorem wholeSpaceInsertion_holds. Any prescribed nonempty open set, with one family and the given reference solution.
Proposition 4.3	Closed. F/Section4/R43/Universal.lean:193, theorem universal_of_memForceR.
Proposition 4.4	Closed. F/Section4/R44/Prop44.lean:24, theorem main. Finite-horizon $L_t^2 H_x^{-1/2}$ estimate in the inhomogeneous norm, with zero initial velocity.
Corollary 4.5	Closed. B/ForceClasses.lean:46, theorem forceClasses.
Proposition 4.6	Closed. B/CompletedDensity.lean:14, theorem completedDensity.
Theorem 4.7	Closed. B/GridObservations.lean:47, theorem gridObservations_choose. Observation times are strictly below the terminal time.

Upstream building block. Under `vendor/NavierStokesAndEuler/NavierStokes/`, `R3ActualCandidate.lean:18` proves `NavierStokes.R3CompactCandidate.selected_compact_candidate`; `R3FiniteEnergyComparison.lean:37` proves `NavierStokes.ComparatorBridge.compact_candidate_excludes_global_solution`. In `ComparatorR3Theorem.lean:36`, `NavierStokes.ComparatorBridge.navier_stokes_breakdown_R3` combines the candidate, uniqueness comparison and viscosity rescaling. `Source/PackageBreakdown.lean` applies positive-viscosity uniqueness to the selected construction and proves the complete statement of Theorem 1.1 for the same force.

The unnumbered analytic inputs are indexed with literature sources and proof locations in Section 4.

3 Differences in proof methods

This section compares the proof routes used in the cited literature with the routes implemented in the code. The coverage of each article statement is listed in Section 2.

Local theory and continuation (Proposition 2.1). Tao's Theorems 5.1 and 5.4 [1] construct local solutions by a contraction in the H^1 mild-solution space and then propagate higher regularity on the same interval. The formalization instead uses its available H^7 local construction on $\mathbb{R}^3$ and H^3 construction on the torus when restarting a smooth solution. To obtain continuation from $\int_0^S \|u(t)\|_{H^2}^2 dt < \infty$, it first applies the high-order energy estimate and Grönwall:

$$\sup_{t < S} \|u(t)\|_{H^m} \leq \left(\|a\|_{H^m} + \int_0^S \|f(t)\|_{H^m} dt \right) \exp \left(C_{m,\nu} \int_0^S \|u(t)\|_{H^2}^2 dt \right) < \infty.$$

Taking $m = 7$ or 3 gives a bounded set of restart data. For the fixed smooth force on a compact time window, the code obtains one positive lifespan valid for all these restart data

and times, then pastes by uniqueness across S . The general H^1 -uniform restart statement is not proved; the continuation argument above uses only the proved high-order case. Tao’s higher-regularity argument (Theorem 5.4(iv), p. 53) explains why smooth data remain in all Sobolev orders; the uniform high-order restart bound is proved separately in the code. See `F/Section4/A04/RestartFixedForce.lean:41`, theorem `restartFixedForce_of_memForceR`, `F/Section4/A04/ShiftedExtension.lean:247`, theorem `extendsBeyond_of_memForceR`’ and `F/Section3/T11/ExistenceInputH3.lean:456`, theorem `extendsBeyondH3`.

Fractional localization (Lemma 3.2). The article invokes Taylor’s coordinate-cover and partition-of-unity construction of Sobolev spaces, with interpolation [2, Chapter 4, Sections 2–3]. A fixed chart cutoff then gives the required bounded local-to-global map. The formal proof establishes the estimate directly for the torus: it identifies homogeneous fractional norms with double-integral seminorms on $\mathbb{R}^3$ and on the torus, compares the periodized kernel with the Euclidean kernel, and bounds the remaining lattice tail by a multiple of $\|z\|_2^2$. The fixed distance from the support ball to the chart boundary makes the constant uniform as the support shrinks. The endpoint orders 0 and 1 are treated separately. The kernel comparison is `F/Section3/T13/KernelComparison.lean:394`, theorem `iTorus_periodize_le`; the resulting estimate is `F/Section3/T13/Assembly.lean:330`, theorem `localizationAPI`. Includes the interior-order estimate and the two endpoint identities.

Critical regularity and non-density. The article invokes the forced critical fixed-point theory [8, Theorem 2.3.1 and Lemma 2.3.2, pp. 47–49], with the inhomogeneous heat estimate (Theorem 2.2.5, p. 44) for the finite-horizon $L_t^2 H_x^{-1/2}$ case. The code instead proves the critical energy inequalities, closes a smallness bootstrap and uses the high-order continuation route above. See the proof locations for Propositions 3.8, 4.3 and 4.4 in Section 2. The open-ball argument yielding non-density is unchanged; only the analytic proof route differs.

4 Classical results available for future reference

The following classical results have formal proofs in this project and may provide starting points for later formalizations. The entries give actual proof locations using the prefixes from Section 2. Each entry gives a literature locator and identifies any specialization or derivation needed to reach the formal version. Unless stated otherwise, the spatial domain is $\mathbb{R}^3$ and vector fields take values in $\mathbb{R}^3$.

1. **Sobolev embedding into bounded functions.** For smooth fields with square-integrable derivatives through order two, the code proves the pointwise and essential-supremum H^2 bounds: `F/Section4/A03/BoundedRepresentative.lean:165`, theorem `enorm_le_jetENorm` and `F/Section4/A03/BoundedRepresentative.lean:182`, theorem `eLpNormTop_le_jetENorm`. These use the derivative-based norm `jetENorm`. *Source:* Tao [3, Appendix A, (A.13), draft p. 336], with $d = 3$, $s = 2$. Apply the scalar estimate componentwise and use Plancherel to pass to the equivalent derivative norm; the literal constant is not claimed to coincide with the code’s constant.
2. **Sobolev algebra and tame product estimates.** At integer orders $m \geq 2$, scalar products satisfy the H^m algebra estimate, and scalar–vector products satisfy the tame estimate with an H^2 low norm: `F/Section4/A03/ScalarTameProduct.lean:308`, theorem `algebraProductScalar` and `F/Section4/A03/VectorTameProduct.lean:273`, theorem `tameProductVector`. *Source:* Tao [3, Appendix A, Lemma A.8, (A.17)–(A.18), draft p. 338]. Formula (A.18) gives the algebra estimate; (A.17) combined with (A.13) gives the H^2 low norm in dimension three. The vector version follows componentwise.
3. **Integral representations of fractional Sobolev norms.** For $0 < s < 1$, the double-integral seminorm equals an explicit constant times the squared homogeneous Fourier norm:

F/Section3/T13/WholeSpaceIdentity.lean:509, theorem `wholeSpace_identity` for smooth compactly supported fields, and F/Section3/T13/TorusIdentity.lean:1174, theorem `torus_identity` for smooth periodic fields, with the mean removed in the homogeneous norm. *Sources:* Di Nezza–Palatucci–Valdinoci [4, Proposition 3.4, arXiv v3, pp. 16–17] for $\mathbb{R}^3$; Roncal–Stinga [5, arXiv v3, Theorem 1.5, (1.6)–(1.8), p. 3; Fourier definition (1.1), p. 1] for the torus. The periodic energy identity is a consequence of their operator formula: set $\sigma = 2s$, pair with the field and symmetrize the integral. Parseval then gives the squared Fourier norm. Rescale their 2π -periodic torus to period one and adjust Fourier constants; the constant mode contributes zero.

4. **Smooth approximation in Sobolev and Bochner spaces.** For every real s , compactly supported smooth vector fields are dense in the implemented H^s datum space: F/Section4/B01/Spatial.lean:134, theorem `spatialApprox`. For $1 \leq q < \infty$, Sobolev-valued $L^q(0, \infty)$ paths can be approximated by finite sums $\sum_j \phi_j(t)A_j$, with $\phi_j \in C_c^\infty(0, \infty)$: F/Section4/B01/Temporal.lean:170, theorem `temporalApprox`. The temporal theorem alone does not make the spatial data A_j smooth; the spatial approximation supplies that separate step. *Sources:* Taylor [2, Chapter 4, Section 1, Exercise 1, author-chapter p. 5] gives Schwartz density for every real s . Multiplying a Schwartz approximant by $\chi(x/R)$ gives convergence in the Schwartz topology and hence in H^s , yielding compact support; this last step is a deduction, not the exercise’s literal statement. Hunter [6, Appendix 6.A, Proposition 6.29, p. 200] gives the finite separated-sum approximation on $(0, T)$. Truncating the integrable q -th power tail first extends it to $(0, \infty)$ for $q < \infty$.
5. **A local vector potential for a divergence-free field.** On a ball, a smooth time-dependent divergence-free field has a smooth vector potential given by the radial integral: F/Section3/T16/BallPotential.lean:324, theorem `exists_potential_on_ball`. *Source:* Petersen [7, Section 7.1, Lemma 7.1.2, pp. 127–128, and Lemma 7.1.5, p. 129]. Apply the homotopy formula to the closed 2-form $\iota_v(dx_1 \wedge dx_2 \wedge dx_3)$ and the radial contraction $x \mapsto x_0 + \rho(x - x_0)$. Translating its primitive into vector notation gives the code’s radial integral. Smooth dependence on time follows by differentiation under the integral over $0 \leq \rho \leq 1$. This specializes the Poincaré lemma to a ball, rather than arbitrary domains.
6. **Classical Navier–Stokes local theory and continuation.** Periodic local existence is proved in F/Section3/T11/ExistenceInputH3.lean:376, theorem `exists_periodicLocalSolution_unconditional`. Whole-space velocity uniqueness and pressure uniqueness up to a time-dependent constant are in F/Section4/A02/Uniqueness.lean:120, theorem `velocity_unique` and F/Section4/A02/Uniqueness.lean:244, theorem `pressure_gauge`. The continuation theorems are listed in Section 3. These results use the project’s smooth initial-data and force classes. *Sources:* Tao [1, Theorems 5.1(ii)–(iv), p. 48, and 5.4(ii)–(iv), pp. 52–53] for local existence, uniqueness and smoothness; Lemma 4.1(i), (40), pp. 44–45, for the whole-space pressure gauge; Corollary 5.2, p. 50, for periodic maximal development and its H^1 blowup alternative. These are the classical local-theory inputs. The precise squared- H^2 continuation test and fixed-force H^7/H^3 restart used here are derived by the high-order estimate and Grönwall in Section 3; they are not quoted as literal conclusions of Tao’s theorems.

References

- [1] T. Tao. Localisation and compactness properties of the Navier–Stokes global regularity problem. *Analysis & PDE* **6**(1) (2013), 25–107. doi:10.2140/apde.2013.6.25.
- [2] M. E. Taylor. *Partial Differential Equations I: Basic Theory*. 2nd ed., Applied Mathematical Sciences, vol. 115. Springer, 2011. doi:10.1007/978-1-4419-7055-8.
- [3] T. Tao. *Nonlinear Dispersive Equations: Local and Global Analysis*. CBMS Regional Conference Series in Mathematics, vol. 106. American Mathematical Society, 2006. doi:10.1090/cbms/106.

- [4] E. Di Nezza, G. Palatucci and E. Valdinoci. Hitchhiker's guide to the fractional Sobolev spaces. *Bulletin des Sciences Mathématiques* **136**(5) (2012), 521–573. doi:10.1016/j.bulsci.2011.12.004.
- [5] L. Roncal and P. R. Stinga. Fractional Laplacian on the torus. *Communications in Contemporary Mathematics* **18**(3) (2016), 1550033. doi:10.1142/S0219199715500339.
- [6] J. K. Hunter. *Notes on Partial Differential Equations*. University of California, Davis, 2014. https://www.math.ucdavis.edu/~hunter/pdes/pde_notes.pdf.
- [7] P. Petersen. *Manifolds, Transversality, and de Rham Cohomology*. University of California, Los Angeles, lecture notes. <https://www.math.ucla.edu/~petersen/manifolds.pdf>.
- [8] R. Danchin. *Fourier Analysis Methods for PDE's*. Lecture notes, 2005. <https://perso.math.u-pem.fr/danchin.rafael/cours/courschine.pdf>.